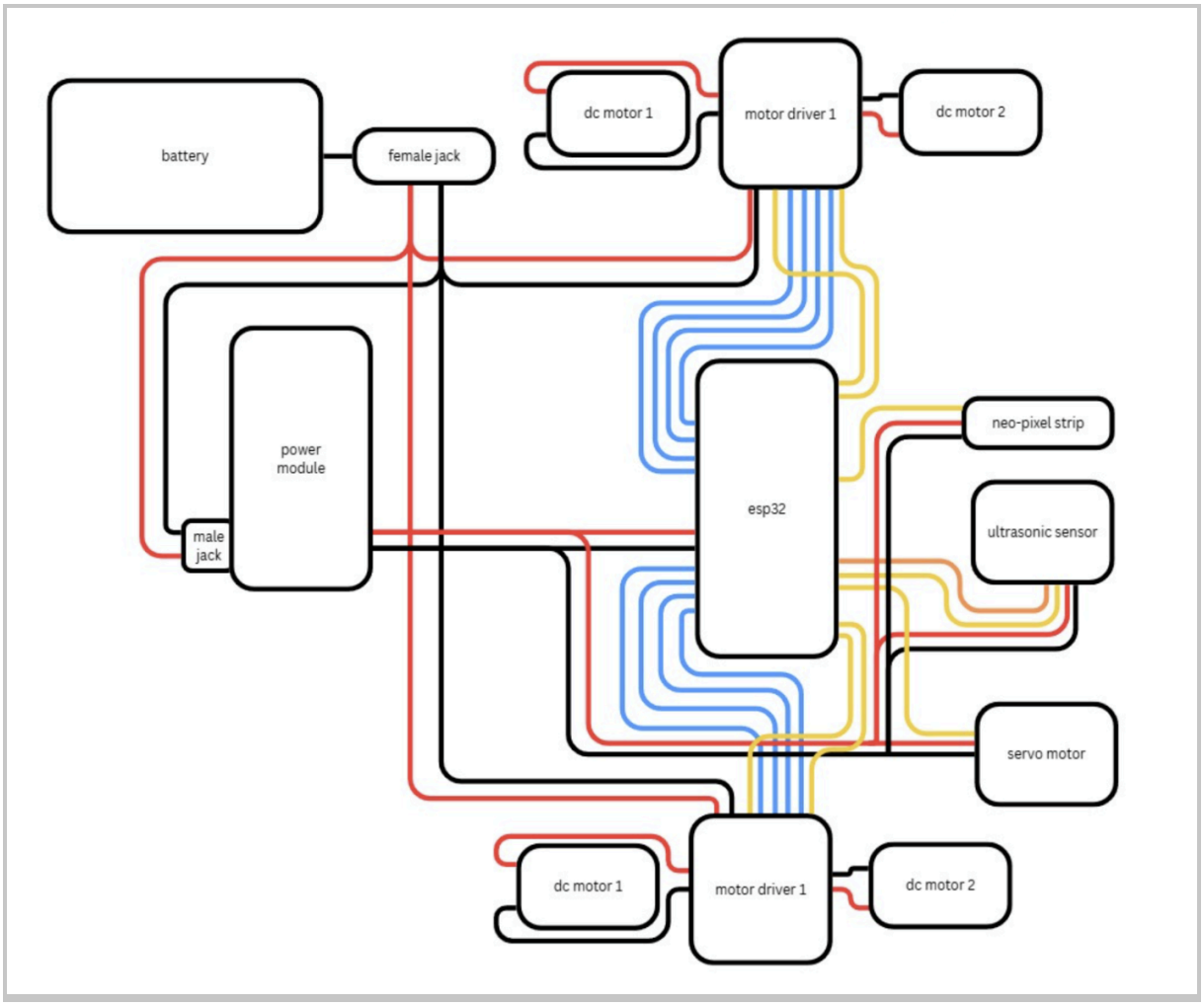
Electronics Documentation

1. Electronics Block Diagram



2. Components Table

Component	Quantity	Purpose	Justification
DC Motor	4	Drives the rotation of each wheel, enabling the vehicle to move.	Can provide continuous rotational motion and allows variable speed control through PWM signals.
Servo Motor	1	Physically rotates the ultrasonic sensor	Rotates to precise, controlled angles with the ultrasonic sensor horizontally across a range of 10° to 170°.
Ultrasonic Sensor	1	Detects obstacles around the vehicle	Facilitates obstacle detection and allows distance measurement without physical contact.
Neopixel Strip	1	Serves as a visual status indicator for the vehicle	Provides a visual feedback system. Where each colour corresponds to a specific state making it easy to monitor the car's behaviour.
ESP32	1	The microcontroller that processes and executes the code	Integrates Wi-Fi and Bluetooth capabilities, processes the code and manages all inputs and outputs across the vehicle.
Power Supply Module	1	Regulates and distributes a stable 5V supply to the components that require it.	Ensures the NeoPixel strip, ultrasonic sensor, and servo motor each receive 5V as per their operating requirements
Physical Switch	1	Manually breaks the circuit to power the vehicle on and off.	Cuts power instantly, preventing unnecessary battery drain, allowing safe handling of the vehicle without disconnecting the wires.
Power rail	1	Distributes the GND and 5V connections to multiple components simultaneously.	A breadboard would not fit in the vehicle, making a power rail the most practical solution for sharing power connections.
Battery	1	Provides the power source supplying the voltage required to the DC motors.	The power supply module provides 5V, DC motors require 6V, this battery can output between 3V to 7.4V, meeting the motors' demands.